

5J and 5K - Review



Year 8 -
Revision of...

Year 8 - Revision of Index Laws, Zero power and Powers of Powers

Wednesday 20th May

Learning Objective: To consolidate understanding of index laws, the zero power, powers of powers/products & quotients

Today we are learning: To be able to evaluate and simplify expressions in which the index forms are multiplied, divided or zero. To be able to Simplify expressions involving powers of powers, products and quotients.

Indicators of progress - I can:

- Evaluate and simplify index forms through multiplication, division
- Apply the index laws regarding zero powers and power of powers.

Which Rule? Game



$$(4^2)^3$$

$$5^4 \div 5^3$$

$$6^4 \times 6^7$$

What do we do to the indexes ?

Hold Up:

- 1 finger = **Add**
- 2 fingers = **Subtract**
- 3 fingers = **Multiply**

Index Law Rule Table:

Index Law	What To Do	Example
Multiplying index forms	Add the powers	$2^3 \times 2^4 = 2^{3+4} = 2^7$
Dividing index forms	Subtract the powers	$5^6 \div 5^2 = 5^{6-2} = 5^4$
Zero power	Answer is always 1	$7^0 = 1$, $100^0 = 1$
Power of a power	Multiply the powers	$(3^4)^3 = 3^{4 \times 3} = 3^{12}$

Maze Activity - Individual Task



Instructions:

1. Start at the box marked "START".
2. Simplify or evaluate the expression inside the box.
3. Look at the answers on the paths around that box.
4. Choose the path with the answer that matches your working.
5. Move to the next box.
6. Repeat:
simplify/evaluate → find the matching answer → follow the path.
7. Show your working in your book or on the sheet.
8. If you reach a dead end, go back and check the last expression.
9. Continue until you reach "FINISH".



To simplify:

- An expression involving index numbers, apply the rules for multiplying and dividing them, writing it in a simpler form. The answer was given in **index form**—its actual value is not calculated.

E.g. $4^2 \times 4^5 = 4^{2+5} = 4^7$

To Evaluate:

- An expression involving index numbers, find the value of the answer as an actual number. The answer is given as a **number**.

E.g. $5^6 \div 5^3 = 5^3 = 125$

Lets Practice!



Question 1:

a. $5^2 \times 5^3$

b. $10^4 \div 10^2$

c. $3^5 \div 3^2$

d. 7^0

e. 12^0

g. $(4^2)^3$

h. $(2^3)^2$

i. $6^2 \times 6^1$

j. $9^5 \div 9^4$

k. $(5^2)^2$

l. $8^3 \div 8^3$

You will need your calculator for this task!



Question 2:

a. $7^5 \div 7^2 \times 7$

h. $(3^2)^4$

h. $11^0 \times 11^3$

i. $(6^2)^2 \div 6^3$

$$c. (10^4)^3$$

$$d. 4^3 \times 4^2$$

$$e. 9^6 \div 9^2$$

$$f. (2^2)^3 \times 2$$

$$g. 5^4 \div 5^1$$

$$j. 8^3 \div 8^4 \div 8$$

$$k. (7^1)^5$$

$$l. 3^3 \times 3^3$$

$$m. (9^2)^2 \div 9^3$$

$$n. 12^4 \div 12^4$$

Question 3: Simplify

$$a. 5^6 \div 5^2 \times 5^3$$

$$b. (10^2)^3 \div 10^4$$

$$c. (3^2)^3 \div 3^4$$

$$d. 7^5 \times 7^2 \div 7^3$$

$$e. (4^3)^2 \div 4^5$$

$$f. 9^4 \div 9^2 \times 9^3$$

$$g. (2^4)^2 \div 2^5$$

$$h. 6^0 \times 6^5 \div 6^2$$

$$i. (8^2)^3 \div 8^4$$

$$j. 12^5 \div 12^2 \div 12^2$$

$$k. (5^3)^2 \times 5^0$$

$$l. 2^7 \div 2^3 \times 2^2$$

$$m. (9^2)^2 \div 9 \times 9^3$$

$$n. (7^4)^2 \div 7^5 \div 7$$

Challenge

Question 4: Simplify

$$a. 5^7 \div (5^2)^2$$

$$b. (3^2)^4 \div 3^5 \times 3$$

$$c. (10^3)^2 \div 10^4 \times 10^0$$

$$d. (4^2)^3 \div (4^3)^2$$

$$e. 8^6 \div 8^2 \div 8^3$$

$$f. (6^2)^3 \times 6^2 \div 6^5$$

$$g. 9^5 \div (9^2)^2$$